

Lecture 6: Sensitivity and Robustness

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
sensitivity	känslighet
robustness	robusthet
crossover frequency	skärfrekvensen
phase margin	fasmarginalen
amplitude margin	amplitudmarginalen
bandwidth	bandbredd
resonance peak	resonanstopp
resonant frequency	resonansfrekvens
peak frequency	resonansfrekvens
phase crossover frequency	fasskärfrekvens
model errors	modellfel
relative error	relativt fel
static gain	statisk förstärkning
loop gain	kretsförstärkning
measurement noise	mätbrus
breakpoints	brytpunkter



Outline

Recap

Introduction

Sensitivity

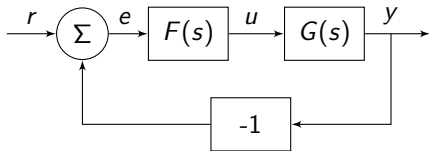
Robustness

End of Part 1

Recap

In the previous lecture..

► Controller design



- Stability
- Servo control problem and steady-state error (determine F so that $y \approx r$)
- Speed and oscillations

► $\Rightarrow$ Approach: Systematic PID/Lead-lag construction.

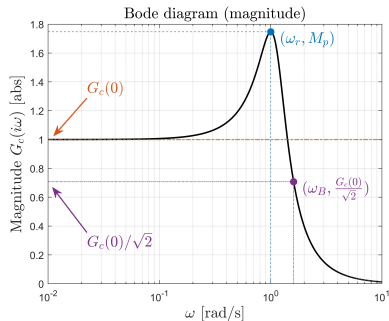
Today:

► Sensitivity and robustness

Recap: Speed and oscillations

From the Bode diagram we have the following quantities:

- ▶ **Bandwidth:*** $|G_c(i\omega_B)| = \frac{1}{\sqrt{2}} = -3dB$:
 $\omega_B \approx$ speed (compare with $1/T_r$)
- ▶ **Resonance frequency:** ω_r
- ▶ **Resonance peak:** $|G_c(i\omega_r)| = M_p \approx$ overshoot M
- ▶ **Steady-state error:** $e_0 = 1 - G_c(0)$



We would like

$$\omega_B \text{ large, } M_p \text{ small, } e_0 = 0$$

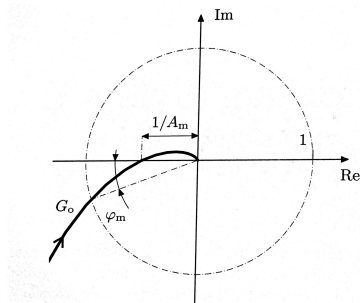
*Note that, if $G_c(0) \neq 1$ then ω_B is s.t. $|G_c(i\omega_B)| \leq |G_c(0)|/\sqrt{2}$

Homework: Study Exempel 5.1!

Recap: The Nyquist plot

- ▶ (gain) crossover frequency / skärfrekvensen ω_c : $|G_o(i\omega_c)| = 1$.
- ▶ phase-crossover frequency / fas-skärfrekvensen ω_p : $\arg(G_o(i\omega_p)) = -180^\circ$.
- ▶ phase margin / fasmarginalen φ_m : $\varphi_m = \arg(G_o(i\omega_c)) - (-180^\circ)$.
- ▶ gain margin / amplitudmarginalen A_m : $A_m = \frac{1}{|G_o(i\omega_p)|}$.

Nyquist criterion: For well-behaved systems, a stability requirement is $\varphi_m > 0$ and $A_m > 1$



$$F(s) = \underbrace{K}_{\text{P}} \underbrace{\frac{\tau_{DS} + 1}{\beta \tau_{DS} + 1}}_{F_{\text{lead}}(s)} \underbrace{\frac{\tau_{IS} + 1}{\tau_{IS} + \gamma}}_{F_{\text{lag}}(s)}$$

- I. speed $\Rightarrow$ requirement on $\omega_c \Rightarrow P$
- II. damping $\Rightarrow$ requirement on $\varphi_m \Rightarrow$ lead
- III. steady-state error $\Rightarrow$ requirement on $|G_o(0)| \Rightarrow$ lag

Recap: Procedure for designing a lead-lag (recipe)

0. Decide which “components” you need (example: do you need a lag?)

P + Lead:

1. Determine the desired $\omega_B \Rightarrow$ Required desired crossover frequency ω_c
2. Determine the desired φ_m (usually $45^\circ - 60^\circ$) $\Rightarrow$ Required phase increase $\varphi_{\max}$ at ω_c
3. Determine β from the diagram of $\varphi_{\max}$ in Figur 5.13 (or calculate it)*
4. From $\omega_c = \frac{1}{\tau_D \sqrt{\beta}}$ (see properties of lead compensator) $\Rightarrow \tau_D = \frac{1}{\sqrt{\beta} \cdot \omega_c}$
5. Choose K such that $|G(i\omega_c)| \cdot \frac{1}{\sqrt{\beta}} \cdot K = 1$, i.e., $K = \frac{\sqrt{\beta}}{|G(i\omega_c)|}$

Lag:

6. Specify $e_0 \Rightarrow$ Calculate γ
7. $\tau_I = \frac{10}{\omega_c}$ (rule of thumb)**. But!, this has a small effect on φ_m since $\arg F_{\text{lag}}(i\omega_c) \geq -5.7^\circ$
Add an additional 5.7° when you consider the phase boost at step 2 of the F_{lead} procedure

***Note!** If a large phase increase is needed, use several compensators in series.

** If the error decreases too slowly $\Rightarrow$ decrease τ_I and compensate for any deterioration of φ_m with F_{lead}



Outline

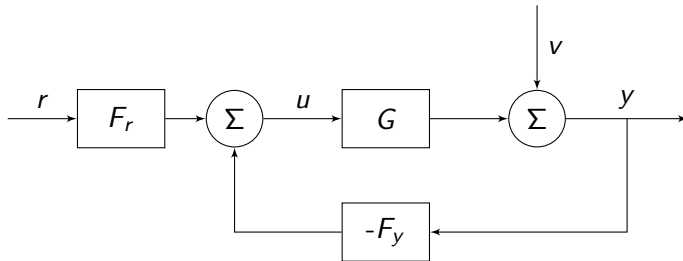
Recap

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End of Part 1



If $F_r = F_y = F \Rightarrow$ we obtain the previous/usual block diagram

Many choices of F_r and F_y give the same closed-loop system $G_c(s)$.

Sensitivity to output disturbance v ?

Example

1. With $F_r = F_y = F$ we obtain

$$\implies G_c = \frac{GF}{1 + GF}$$

2. With open-loop control ($F_y = 0$) and $F_r = \frac{F}{1+GF}$ we obtain

$$\implies G_c = \frac{GF}{1 + GF}$$



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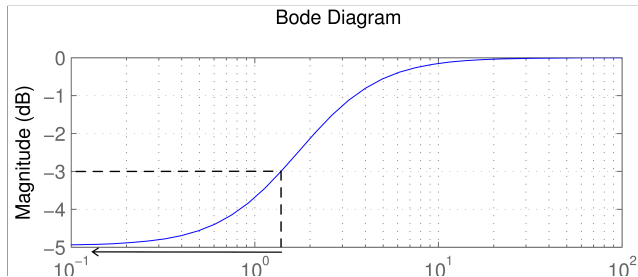
End of Part 1

Benefits/Advantages of feedback:

- Feedback reduces the impact of disturbances

$$Y = \frac{GF_r}{1 + GF_y} R + \underbrace{\frac{1}{1 + GF_y}}_{\text{sensitivity function } S} V \implies S(s) = \frac{1}{1 + G(s)F_y(s)}$$

We obtain a small impact from disturbances at frequencies where $|S(i\omega)|$ is small.



Example (PI control)

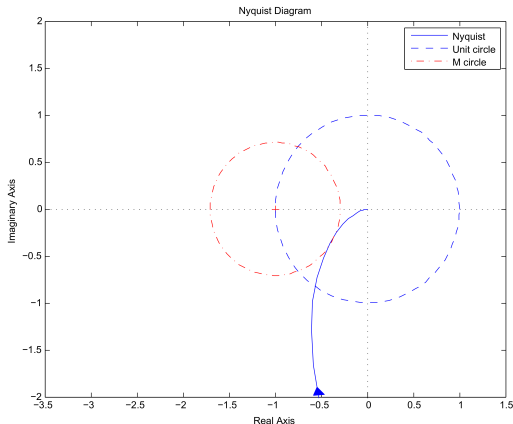
$$\Rightarrow |S(0)| = 0$$

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More precisely

$$|S(i\omega)| = \left| \frac{1}{1 + G(i\omega)F_y(i\omega)} \right| \leq M_s \Leftrightarrow |G(i\omega)F_y(i\omega) - (-1)| \geq \frac{1}{M_s}$$

The Nyquist curve must lie outside a circle with center at -1 and radius $1/M_s$.





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- Feedback reduces the impact of certain types of model errors

Model: $G(s)$

True system (verklighet): $G^0(s) = G(s)(1 + \Delta_G(s))$

We can solve $\Delta_G(s) = \frac{G^0(s) - G(s)}{G(s)}$ and see that it is a *relative error*.

What happens when one closes the loop with the **true** system?

If we feed back the **true** system, we get:

$$Y^0 = \frac{G^0 F_r}{1 + G^0 F_y} R = \dots = (1 + S^0 \Delta_G) Y, \quad \text{where} \quad S^0(s) = \frac{1}{1 + G^0(s) F_y(s)}$$

The relative error in the output signal is:

$$\Delta_Y = \frac{Y^0 - Y}{Y} = S^0 \Delta_G$$

$\Rightarrow$ The model error $\Delta_G(i\omega)$ has little impact
if the **true sensitivity function** $|S^0(i\omega)|$ is small.

Example

PI control $\Rightarrow S^0(0) = 0 \Rightarrow$ Eliminates the impact of incorrect static gain, $G^0(0)$ (robust!).

Problem: Usually, S^0 is not known (it is based on true system G^0)

However, in the case of “small” model error, one can use with good approximation:

$$S^0 \approx S = \frac{1}{1 + GF}$$

Sensitivity and robustness

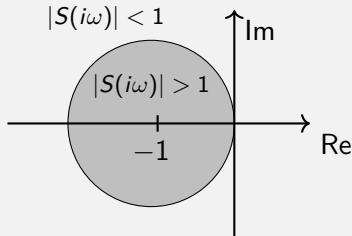
Example

For the case $F_y = F_r = F$ we obtain

$$G_c(s) + S(s) = \underbrace{\frac{G(s)F(s)}{1 + G(s)F(s)}}_{\approx 1} + \underbrace{\frac{1}{1 + G(s)F(s)}}_{\Rightarrow \approx 0} = 1$$

Good choice!

Nyquist plot and $S(i\omega)$ (compare the M_s -circle):





- ▶ High loop gain increases the risk of instability, especially in the event of model errors (robustness)
- ▶ May require large control signals.

A large control signal is required to eliminate interference in frequency bands where $|G(i\omega)|$ is small.

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If G_0 has a relative degree ≥ 2 , then

where $p_1, \dots, p_m$ are the poles of $G_0(s)$ in RHP.

Observe that this is independent of the regulator.

The proof is based on the integration of

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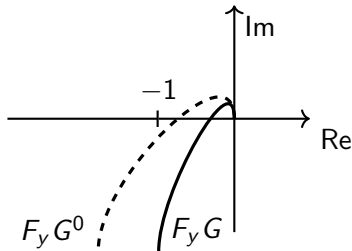
Model: $G(s)$, assumed stable

True system: $G^0(s) = G(s)(1 + \Delta_G(s))$, assumed stable

The nominal closed-loop system: $\frac{GF_r}{1+GF_y}$, assumed stable

Question: Is the true closed-loop system stable?

To answer this question, we can use the Nyquist criterion, among others.



$$F_y G^0 = F_y G(1 + \Delta_G) = F_y G + F_y G \Delta_G$$

The requirement is that $F_y G^0$ must *not* encircle -1 .

OK, if the distance between $F_y G$ and $F_y G^0$ is less than the distance between $F_y G$ and -1 .

This gives the so-called **Robustness criterion** from 1978 [Resultat 6.2]

$$\Rightarrow |F_Y(i\omega)G(i\omega)\Delta_G(i\omega)| < |F_Y(i\omega)G(i\omega) + 1|$$

$$\Rightarrow \frac{|F_y(i\omega)G(i\omega)|}{|1 + F_y(i\omega)G(i\omega)|} < \frac{1}{|\Delta_G(i\omega)|} \quad \forall \omega$$

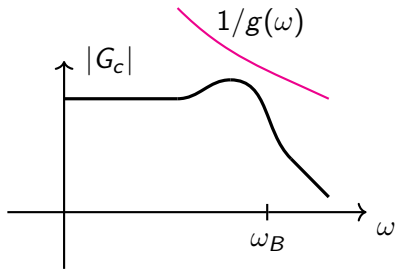
$$T(s) = \frac{F_y(s)G(s)}{1 + F_y(s)G(s)} = 1 - S(s)$$

Observe: Usually, Δ_G is unknown, but an upper estimate is given

Sufficient condition: The true closed-loop system is then stable if

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Observe that if $F_y = F_r = F$ then it is $T = G_c$.



Limits the available/possible bandwidth.

Example

Assume that

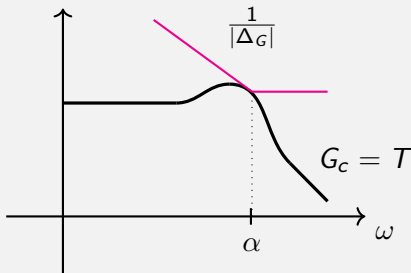
$$G^0(s) = G(s) \underbrace{\frac{\alpha}{s + \alpha}}_{1 + \Delta_G(s)} \Rightarrow \Delta_G(s) = \frac{\alpha}{s + \alpha} - 1 = -\frac{s}{s + \alpha}$$

Then: $\frac{1}{\Delta_G(s)} = -\frac{s + \alpha}{s}$

Its Bode plot has breakpoints at 0 (slope -1) and α (slope $+1$).

The slower the neglected dynamics (α), the lower the possible bandwidth.

Alternatively: Model better!





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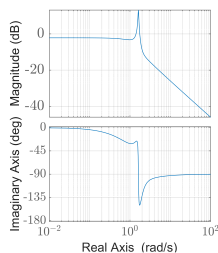
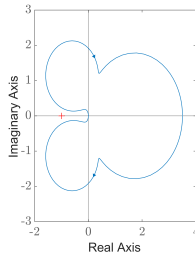
Sensitivity

Robustness

End of Part 1

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We have now completed the first part of the course
(classical control theory): Laplace, Nyquist, Bode, etc.



Next time: modern control theory (1960)!

